

Social Media Analysis of the Ferrari Luce

Arseniy Kirykouski* Michele Pio Lacagnina[†]

Stefano Rizzuto[‡]

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1 Introduction and Motivation

This study investigates the global online response to the May 25, 2026 launch of the Ferrari Luce, the manufacturer’s inaugural production battery-electric vehicle (BEV) and first five-seat model. The vehicle’s debut lead to a highly polarized public discourse. The multifaceted nature of this launch, including the brand’s pivot to electrification, notable design collaborations (e.g., Jony Ive, Marc Newson, and LoveFrom), and the disruption of established brand heritage provides a robust paradigm for analyzing intersecting digital subcultures. Specifically, the event brought engagement across diverse stakeholder groups, including automotive purists, electromobility advocates, design enthusiasts, motorsport followers, and financial investors. As a result, the primary objective of this research is to empirically characterize the online discourse surrounding the Ferrari Luce. We aim to model the conversational network, quantify emotional and aspect-level sentiments, identify drivers of the observed public backlash, and trace the temporal evolution of these dynamics, particularly in relation to the concurrent decline in stock valuation.

1. **Network Topology:** Which nodes demonstrate the highest network centrality (e.g., influencers, brokers, bridge accounts), and does the conversational structure exhibit a centralized hub or a fragmented network topology?
2. **Community Dynamics:** What distinct discourse communities emerge, what is their structural cohesion (modularity), and to what extent do contrasting ideological factions interconnect or isolate into echo chambers (assortativity)?
3. **Temporal Sentiment & Emotion:** What is the aggregate affective polarity and emotional distribution, and how do these metrics vary across

*arseniy.kirykouski01@universitadipavia.it

[†]michelepio.lacagnina01@universitadipavia.it

[‡]stefanoanthony.rizzuto01@universitadipavia.it

communities, particularly between the initial vehicle reveal and the subsequent stock market drop?

4. **Aspect-Based Sentiment:** Which specific product dimensions (e.g., exterior/interior aesthetics, acoustic engineering, price-to-value ratio, performance metrics, nomenclature, electrification strategy, brand heritage, and the LoveFrom design influence) serve as the primary factors for negative sentiment?
5. **Entity & Comparative Analysis:** Which external entities and comparative benchmarks (e.g., Nissan Leaf, Tesla, Apple, CEO Benedetto Vigna, regulatory figures) emerge most frequently within the discourse, and what is the sentiment polarity associated with each?
6. **Linguistic Nuance:** What is the prevalence of sarcasm and irony within the critical discourse, and to what degree do these linguistic features introduce distortion into baseline sentiment classification models?
7. **Stance Detection:** Independent of overt tone, what is the underlying stance of the discourse regarding the Ferrari Luce specifically, and the broader macroeconomic transition to electric vehicles generally (e.g., isolating users who are pro-EV but anti-Luce)?

2 Data description

2.1 Data Collection

The data for this project came from two main platforms: Bluesky and Reddit.

For the Bluesky data, we used the AT Protocol API to find posts about the Ferrari Luce. For every post, the system saves the text, author details, time of posting, engagement numbers, hashtags, mentions, links, language tags, and thread information. It also has an option to download the follower networks for specific accounts.

For Reddit, the script prefers using PRAW, which is the official Reddit API. If API keys are not provided, it uses Arctic-Shift and PullPush.io as backups. We collected posts and comment threads from relevant subreddits like `r/Ferrari`, `r/cars`, `r/electricvehicles`, `r/formula1`, `r/stocks`, and `r/wallstreetbets`.

We also applied a relevance filter to the Reddit data. This is an important step because “luce” means “light” in Italian. To avoid collecting unrelated posts, the algorithm only keeps text that contains the exact phrase “Ferrari Luce” or the word “Ferrari” alongside EV-related terms.

1 shows examples of raw records collected from both platforms before they were combined into the main dataset. These examples highlight the variety in our data: short opinion posts, longer Reddit submissions, multilingual content, and direct comment replies. Bluesky posts mainly provide public reactions, hashtags, language tags, and engagement metadata. Meanwhile, Reddit records

include both submissions and threaded comments, allowing us to map out the back-and-forth interactions between users. These excerpts also introduce the main themes we analyzed later in the project, such as design criticism, price comparisons, Jony Ive’s role, brand identity, and polarized user debates.

Platform	Type	ID / URI	Date	Source	Raw text excerpt	Why it is useful
Bluesky	Post	at://.../3mnk aupiy2s2y	2026-06-05	Bluesky	"La Ferrari Luce ha un design anonimo e costa 500.000€, il triplo di una Tesla..."	Italian-language criticism mentioning design, price, and Tesla comparison
Bluesky	Post	at://.../3mnj fvoxc2t	2026-06-05	Bluesky	"...people calling in to say the ferrari luce looks fine... it's a bad looking car..."	English opinion post focused on exterior design
Reddit	Submission	1twpt2u	2026-06-04	r/Ferrari	"Jony Ive Shows Me The Most Controversial Ferrari Ever (Ferrari Luce)..."	Shows discussion around Jony Ive/design influence
Reddit	Submission	1tunlve	2026-06-02	r/Ferrari	"Can we please stop with the AI design posts?... majority of us don't like the Luce direction..."	Long discussion post about design backlash and community behavior
Reddit	Comment	opw23j0	2026-06-05	r/Ferrari	"...any changes at this stage will be incredibly costly and very likely superficial."	Example of practical/industry-oriented discussion
Reddit	Comment	opvhd8o	2026-06-05	r/Ferrari	"So you complain about 'car guys' in a 'car guys' subreddit..."	Example of conversational reply structure used for network edges

Table 1: Examples of raw records collected from Bluesky and Reddit.

2.2 Corpus Construction and Cleaning

After collecting the raw data, we merged the Bluesky posts, Reddit submissions, and Reddit comments into one main dataset. This table uses a standard format: `[doc_id, source, author, created_at, text, lang]`. To clean the data, the code removes URLs, fixes extra spaces, uses `langdetect` to fill in missing language tags, and filters out known bot or moderator accounts. This gives us a refined dataset made up of real user conversations.

An important part of our method is that we process the text in two different ways depending on what it will be used for:

- **A word-level path for lexicon methods:** This uses the NLTK Tweet-Tokenizer, removes stopwords, and applies lemmatization.
- **A transformer path for neural models:** This keeps the original social media text mostly the same and only normalizes mentions and URLs.

This approach works because lexicon methods need clean, individual words to work properly, while transformer models perform best when the text looks similar to the natural language they were originally trained on.

3 Methodology and Results

3.1 Network Construction

We built the interaction network as a **directed, weighted graph**. In this model, each node represents a social media account, and each edge shows an interaction between two accounts. For Bluesky, edges were created from mentions and replies. For Reddit, edges were based on the comment threads, connecting a comment’s author to the author of the original post or the parent comment.

The direction of the edge shows the flow of attention: an edge pointing from User A to User B means that User A interacted with User B. If they interacted multiple times, the edge weight increases.

After removing bots and deleted accounts, the final graph had **8,223 nodes** and **13,792 directed edges**. Out of these nodes, **8,018 were Reddit accounts** and **205 were Bluesky accounts**, which shows that the conversation was mostly dominated by Reddit. The network is very sparse, with a density of **0.00020**. This means only a tiny fraction of all possible connections actually exist, which is normal for social media networks where users usually only talk to a few people rather than the whole group. The graph also includes **125 weakly connected components** and **6,149 strongly connected components**. This tells us the conversation was fragmented into many separate threads instead of being one large, connected public debate.

The reciprocity score is **0.2338**, meaning only about a quarter of the interactions were mutual. This makes sense for comment sections, where people often leave a reply without getting an answer back. Furthermore, the low transitivity value (**0.0050**) and average clustering coefficient (**0.0368**) show that users did not form dense groups where everyone talks to everyone else.

Finally, we calculated centrality measures to find the most important accounts in the structure. In-degree and PageRank highlight users who received the most attention, out-degree highlights the most active users, and betweenness points out users who acted as bridges between different discussions. In short, these numbers show that the Ferrari Luce conversation was not one united community, but rather a collection of loose threads with a few key users who drew attention or connected different parts of the discussion. The data are summarized in table 2

Metric	Value	Interpretation
Nodes	8,223	Accounts included in the interaction graph
Edges	13,792	Directed interactions between accounts
Density	0.00020	Very sparse network
Transitivity	0.0050	Very low triangle closure
Average clustering	0.0368	Few dense local groups
Reciprocity	0.2338	Around one quarter of interactions are mutual
Weakly connected components	125	Conversation split into many disconnected areas
Strongly connected components	6,149	Few mutually reachable directed groups
Degree assortativity	-0.1590	High-degree users tend to connect with lower-degree users

Table 2: Interaction Graph Network Metrics

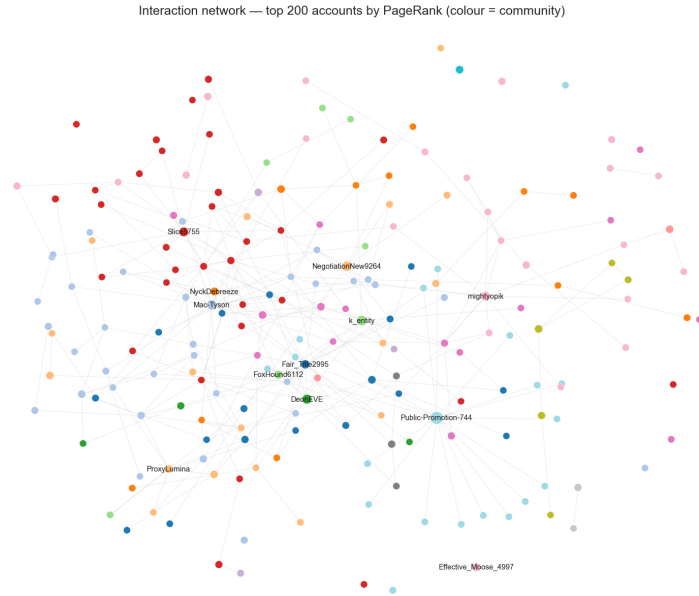


Figure 1: Interaction network showing the top 200 accounts by PageRank.

3.2 Centrality Analysis

Once the directed interaction graph was built, we calculated centrality measures to find the most important users in the Ferrari Luce conversation. Because the edges represent directed interactions like replies or mentions, the in-degree shows how much attention an account received, while the out-degree shows how actively that account interacted with others. We used PageRank as our primary measure of influence because it rewards users who are connected to other important accounts, rather than just those with a high number of basic interactions.

The highest PageRank score belonged to the user **Public-Promotion-744** with a value of 0.0251. This was significantly higher than the second-ranked account, **k_entity** (0.0128), which suggests that the first user held the most influential position in the network. Furthermore, all of the top-ranked accounts are from Reddit, confirming that the core of the interaction network was driven by Reddit comments rather than activity on Bluesky.

Interestingly, some users like **DeonEVE** and **k_entity** have very high in-degree scores (363 and 260, respectively) but an out-degree of zero. This means they received a lot of attention but did not actively reply to others within our dataset. On the other hand, the user **Fair_Title2995** combined a high in-degree (173) with the highest out-degree among the top PageRank users (81). They also had the highest betweenness centrality (0.0598), pointing to a more active role in bridging different parts of the discussion.

Overall, these centrality results show that influence in this network is not just about being popular. It also depends on participating in reply chains and holding positions that connect separate groups of conversation.

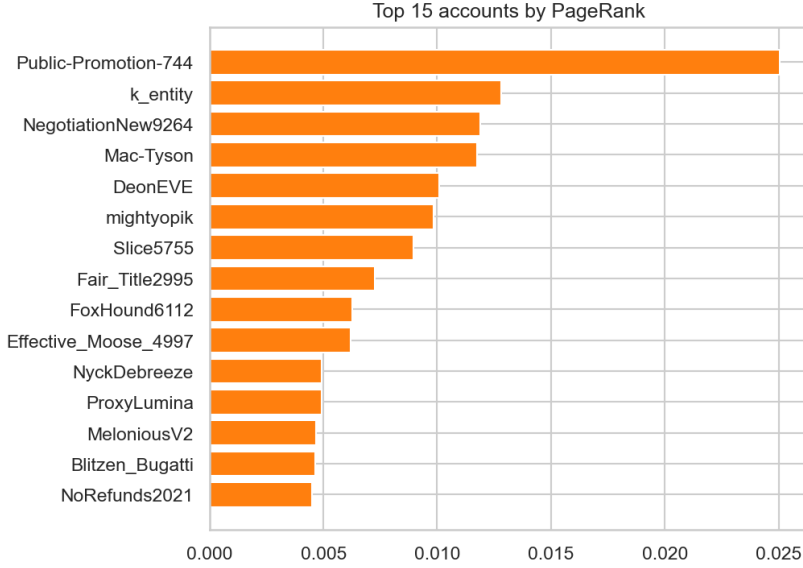


Figure 2: Top 15 accounts by PageRank

Rank	Account	Platform	In-degree	Out-degree	PageRank	Betweenness	Interpretation
1	Public-Promotion-744	Reddit	259	17	0.0251	0.0078	Most influential node by PageRank
2	k_entity	Reddit	260	0	0.0128	0.0000	Highly addressed, passive in collected replies
3	NegotiationNew9264	Reddit	305	6	0.0119	0.0118	Very high attention and some bridging
4	Mac-Tyson	Reddit	252	12	0.0118	0.0175	Influential and structurally connective
5	DeonEVE	Reddit	363	0	0.0101	0.0000	Highest in-degree, mostly attention recipient
6	mightyopik	Reddit	142	12	0.0098	0.0097	Moderately influential bridge
7	Slice5755	Reddit	38	7	0.0089	0.0042	Lower degree but relatively strong PageRank
8	Fair_Title2995	Reddit	173	81	0.0073	0.0598	Main broker / bridge among top users
9	FoxHound6112	Reddit	256	1	0.0063	0.0000	High attention, little outward interaction
10	Effective_Moose_4997	Reddit	64	20	0.0062	0.0042	Active participant with moderate brokerage

Table 3: Top 10 Accounts by PageRank in the Interaction Graph

3.3 Community Detection

To perform community detection, the directed interaction network is first converted into an undirected weighted graph. Consequently, repeated or reciprocal interactions are merged into undirected edges, and their corresponding weights are aggregated.

Two modularity-based algorithms are applied to this undirected weighted graph: Louvain and greedy modularity. The Louvain algorithm serves as the primary approach, while greedy modularity provides a baseline for comparison. The Louvain method identified **161 communities** with a modularity score of **0.7600**, whereas the greedy modularity algorithm detected **166 communities** with a score of **0.7556**. Because the Louvain method achieved a slightly higher modularity, its partition is preferred. A modularity score of approximately **0.76**

is considered high, indicating that users interact significantly more within their designated groups than with external users. This implies that the Ferrari Luce discussion consists of distinct, separated discourse groups rather than a single homogeneous conversation. This is consistent with the fact that users tend to comment mostly in one subreddit of interest.

Following detection, each user is assigned two community labels corresponding to the Louvain and greedy modularity results. The subsequent analysis primarily utilizes the Louvain labels. To ensure that isolated or minor clusters do not skew the interpretation, small communities with fewer than **5 users** are consolidated into a single ‘other’ category. After applying this filter, the Louvain partition consists of **49 communities with at least 5 members**, while **255 accounts** are categorized within the ‘other’ group.

The largest identified communities are relatively big. Specifically, community **125** contains **417 members**, community **92** contains **411 members**. These values demonstrate that the network comprises multiple large conversational clusters rather than being dominated by a single group (table 4).

Community names are assigned after the detection phase. We profile each community using the posts written by its members. First, we identify the dominant subreddit associated with the users and documents in a given community. If at least **60%** of the community’s mapped documents originate from the same subreddit, we assign it a human-readable base label, such as **Ferrari faithful**, **Car enthusiasts**, **EV enthusiasts**, **F1 fans**, or **Traders / investors**. If no subreddit reaches this threshold, the community is labeled **mixed / other**.

Furthermore, we extract distinctive keywords for each community using TF-IDF. These keywords are used to refine certain labels with specific themes, such as **Apple/design**, **styling backlash**, **purist/identity**, or **price/market**, provided the community strongly over-indexes on those terms. Finally, we consolidate communities with similar base labels into broader discourse camps. This structure allows to compare broader audience groups.

By connecting back these communities with their source subreddits and textual content, the principal discourse camps become clearly identifiable. The largest camp is the **Ferrari faithful**, comprising **15 communities**, **3,617 members**, and **7,554 documents**. This highlights that Ferrari-oriented users account for the majority of the discussion volume. The subsequent largest groups are summarized in table 5

Table 4: Community Detection Metrics and Interpretations

Metric	Result	Interpretation
Louvain communities	161	The interaction network is split into many distinct user groups.
Louvain modularity	0.7600	High modularity; communities are well separated and internally dense.
Greedy communities	166	A second algorithm finds a similar number of groups.
Greedy modularity	0.7556	Slightly lower than Louvain, supporting Louvain as the main partition.
Minimum community size	5 users	Groups smaller than this are not interpreted individually.
Louvain communities after filtering	49	These are the communities large enough to be analyzed.
Accounts in “other”	255	Small or isolated clusters are merged to avoid overinterpretation.
Documents mapped to communities	16,602 / 19,840	Most documents could be linked to an author community.

Table 5: Discourse Camps Overview

Camp	Communities	Members	Docs
Ferrari faithful	15	3,617	7,554
Car enthusiasts	7	1,533	3,556
EV enthusiasts	3	1,081	2,196
Mixed / other	6	618	1,211
F1 fans	5	779	1,179
Traders / investors	4	580	831

Community structure — top 10 communities, every user equal size (PageRank-independent). Distinct communities (#id) may share a category label.



Figure 3: Communities visualization

3.4 Sentiment Analysis

Building upon the data processing pipelines established earlier, we compute both lexicon-based and transformer-based sentiment. **VADER** and **AFINN** serve as interpretable baseline methods, evaluating words and expressions against predefined sentiment dictionaries. Specifically, **VADER** generates a continuous `vader.compound` score ranging from negative to positive. We then convert this

continuous score into discrete labels using specific thresholds: values below **-0.05** are classified as negative, values between **-0.05** and **0.05** are neutral, and values above **0.05** are positive.

Conversely, we employ a transformer model, specifically cardiffnlp/twitter-roberta-base-sentiment-latest, as our primary sentiment classifier because it is significantly better adapted to social media language and contextual meaning. Our main analysis relies on the transformer labels.

Table 6: Comparison of Sentiment Analysis Methods

Method	Positive	Neutral	Negative
VADER	9,149 (46.1%)	5,781 (29.1%)	4,910 (24.7%)
AFINN	8,305 (41.9%)	6,827 (34.4%)	4,708 (23.7%)
cardiffnlp/twitter-roberta-base-sentiment-latest	3,403 (17.2%)	8,938 (45.1%)	7,499 (37.8%)

As shown in Table 6, the lexicon-based methods generally yield more optimistic results. Specifically, **VADER** gives the most positive reading of the discussion, likely reacting strongly to isolated positive words (such as “beautiful,” “fast,” or “Ferrari”) and enthusiasm markers, even when the overall context of a comment is skeptical or ironic. **AFINN** produces a similarly positive distribution, though it assigns more neutral scores because it only evaluates words present within its specific dictionary. In contrast, the transformer model provides a much more cautious and critical assessment. It detects significantly more negative and neutral content, as it is better equipped to capture sentence context, the nuances of social media language, and indirect criticism. The results of the transformer model are also visible in Figure 4

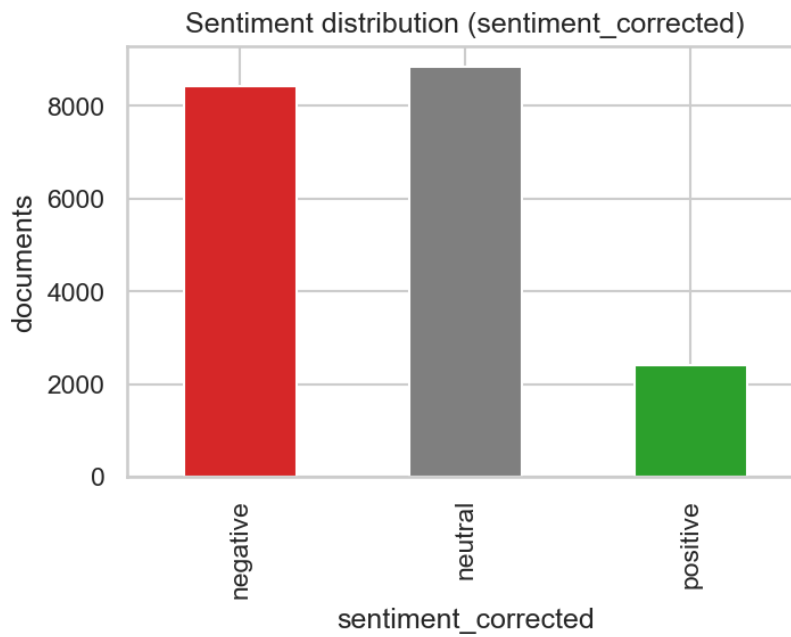


Figure 4: Predicted Twitter-roberta sentiment

The Ferrari Luce discussion is not overwhelmingly positive. The dominant tone is neutral-to-negative, with negativity more than twice the positive share.

We also analyzed each document giving it a sentiment value for eight emotion categories: anger, anticipation, disgust, fear, joy, sadness, surprise, trust using the NRC emotion lexicon. The mean sentiment frequency is visible in 5

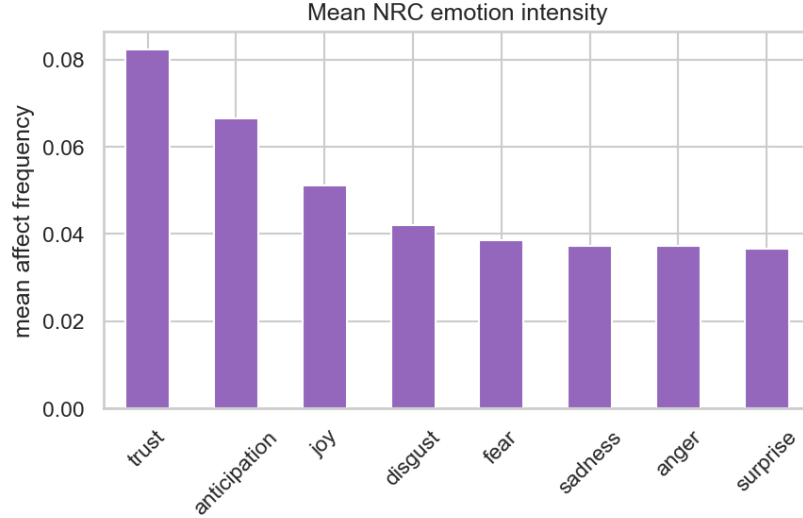


Figure 5: NRC emotion

Since this method uses specific words to assign the sentiment values even short or ironic document might get a high "trust" value. Indeed looking into the data one can see that a comment like: "The price tag and the badge are the only thing about it that makes it a Ferrari. Horrible." got trust value 1 while the transformer labeled it as negative. For this reason we later used the transformer labels as the main sentiment.

3.4.1 Aspect-based Sentiment

As detailed in Table 7, **Electrification** emerges as the most frequently discussed topic, attracting substantial criticism. However, the **Exterior look** (0.471) and **Price** (0.466) represent the most polarizing and negatively perceived product dimensions overall. Similarly, discussions concerning **Brand identity** exhibit high negativity (0.462), with many users interpreting the vehicle as a challenge to Ferrari's traditional heritage. While **Sound** generates a lower volume of discussion, it remains meaningfully negative, likely reflecting broader enthusiast concerns regarding electric vehicles. Conversely, the vehicle's **Name** receives comparatively less criticism (0.248), and the conversation surrounding the **Design house** indicates a mixed reception that is notably less negative than the direct critiques of the exterior styling. Finally, discussions regarding **Performance** and the **Interior** display more moderate levels of negativity.

Table 7: Aspect-Based Sentiment Analysis

Aspect	Total Docs	Negative Ratio
Electrification	7,860	0.420
Exterior look	6,068	0.471
Design house	3,751	0.394
Name	3,284	0.248
Price	2,883	0.466
Performance	1,648	0.351
Interior	1,445	0.365
Brand identity	1,400	0.462
Sound	425	0.412

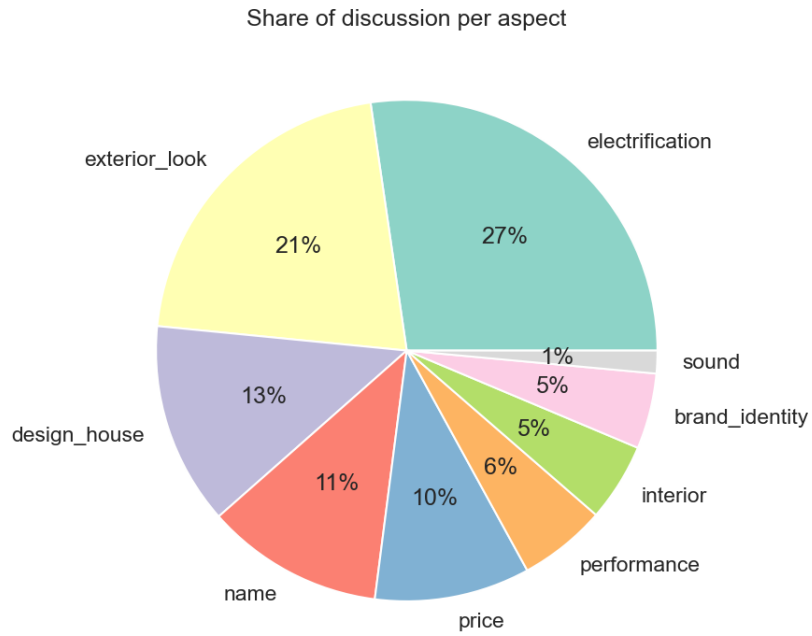


Figure 6: Share of aspects discussed

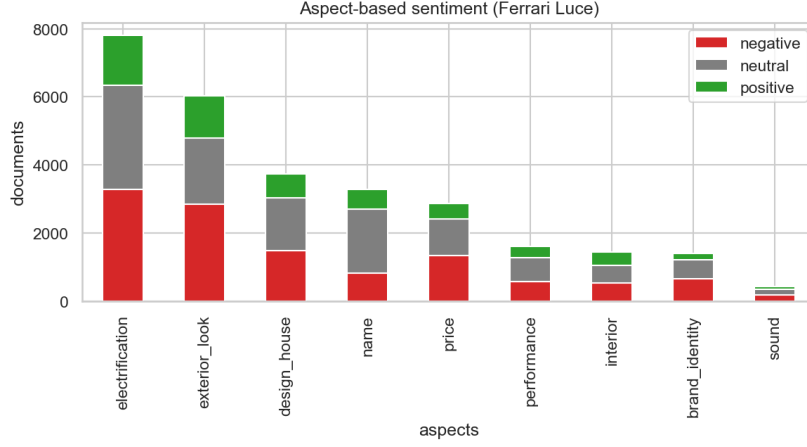


Figure 7: Sentiment per aspect

Negativity is not generic: it concentrates around exterior look, price, and brand identity. This means users are not only reacting to the fact that the car is electric, but also to whether it still “feels like a Ferrari.”

3.4.2 Community-level Sentiment

For each broader community found, the number of positive, neutral, and negative documents is counted and converted into proportions, which is why each horizontal bar sums to 1. The results show that all camps contain a large neutral component, but negative sentiment is consistently larger than positive sentiment. Traders / investors are the most critical group, with about half of their documents classified as negative, while mixed / other and F1 fans show comparatively lower negativity. This suggests that reactions to Ferrari Luce are not uniform across the network: financial and general automotive audiences are more skeptical, whereas F1-oriented and mixed communities express a more moderate tone. This is visible in Figure 9

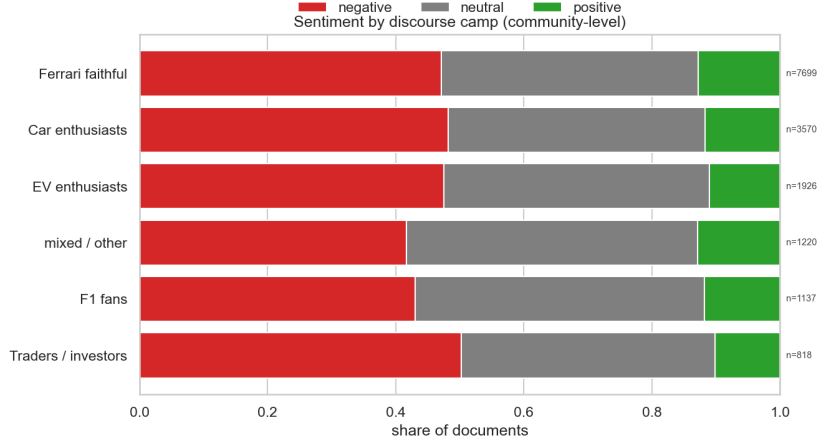


Figure 8: Sentiment per community

3.5 Named Entity Recognition

We applied Named Entity Recognition (NER) using spaCy to identify prominent people, brands, places, products, and organizations within the corpus. This process extracted **6,553** distinct entities. The results indicate that while the discussion is primarily focused on Ferrari and the Luce, it is heavily influenced by comparative references to entities such as Apple, Jony Ive, Porsche, Tesla, Honda, and China.

Furthermore, entity-level sentiment analysis demonstrates that these references vary significantly in tone. For instance, while the specific term “Ferrari Luce” generally appears in neutral contexts, references related to China exhibit high negative ratios. This suggests that comparisons involving Chinese electric vehicles or styling cues are frequently used as a point of criticism.

Finally, co-occurrence metrics reveal three primary interpretive frames within the discourse: discussions focused directly on the Ferrari Luce product, debates surrounding the design influence of Apple and Jony Ive, and comparative analyses involving competitors like Porsche, Tesla, and Honda.

As summarized in Table 8, the core entity metrics confirm that while baseline product terms like **ferrari luce** appear predominantly in neutral contexts with a low negative ratio of **0.132**, external comparative entities invoke significantly more critical sentiment. Specifically, the entity **chinese** exhibits the highest negative ratio at **0.544**, followed closely by **china** (**0.462**), **tesla** (**0.422**), and **bmw** (**0.420**). Conversely, design-related entities such as **apple** (**0.365**), **jony** (**0.303**), and **lovefrom** (**0.253**) show comparatively lower levels of negativity, though they still represent substantial pillars of the overall conversational volume.

Table 8: Entity-Level Sentiment Analysis

Entity	Total Mentions	Positive	Neutral	Negative	Negative Ratio
ferrari	5,829	935	2,638	2,256	0.387
luce	734	168	322	244	0.332
ferrari luce	515	66	381	68	0.132
chinese	379	38	135	206	0.544
china	210	22	91	97	0.462
tesla	211	44	78	89	0.422
bmw	119	22	47	50	0.420
apple	353	75	149	129	0.365
jony	290	77	125	88	0.303
lovefrom	83	23	39	21	0.253

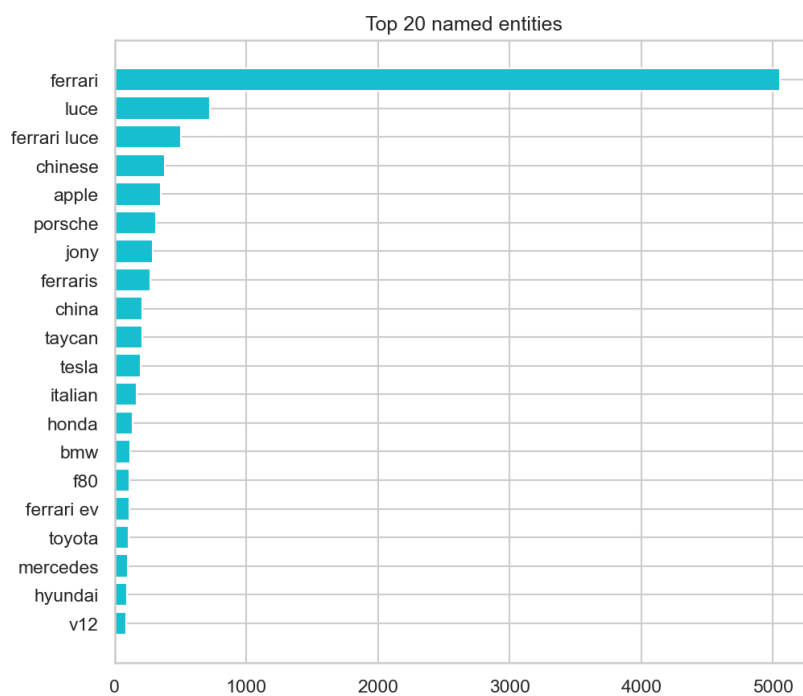


Figure 9: Top 20 most frequent entities

4 Conclusions

To conclude, this project shows that the online reaction to the Ferrari Luce is highly fragmented and only partially positive. The network analysis reveals a sparse but clearly structured interaction space, where users are separated into distinct discourse communities rather than participating in one unified conversation. The high Louvain modularity score of **0.7600** confirms that these groups are internally cohesive and relatively separated from each other. When these communities are linked back to textual content, the main camps that emerge are Ferrari faithful, car enthusiasts, EV enthusiasts, F1 fans, traders/investors, and mixed audiences. This demonstrates that the Ferrari Luce is not interpreted in the same way by all users: each group evaluates the car through its own expectations, interests, and cultural references.

From the content analysis, the overall discourse appears mainly neutral-to-negative. The transformer sentiment model classifies 37.8% of documents as negative and only 17.2% as positive, suggesting that public reception is marked by skepticism more than enthusiasm. Aspect-based sentiment shows that criticism is especially concentrated around **exterior design**, **price**, and **brand identity**, indicating that the controversy is not limited to electrification itself, but also concerns whether the Luce fits the symbolic meaning of Ferrari. The entity and co-occurrence analyses support this interpretation, as frequent references to Apple, Jony Ive, Porsche, Tesla, China, and Honda show that users evaluate the Luce through comparisons with design culture, EV competitors, and perceived brand authenticity. Overall, the findings suggest that the Ferrari Luce launch generated strong visibility and debate, but also exposed a tension between innovation and heritage: while some users recognize the strategic need for electrification, many perceive the model as challenging Ferrari’s traditional identity.